Hyperbolas

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Standard form

For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

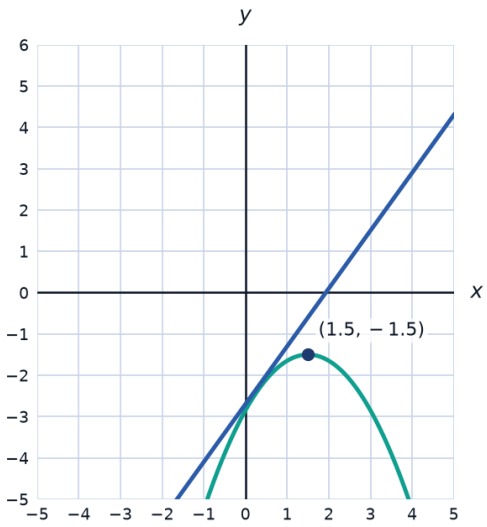
a the vertices

b the foci

c the equations of the asymptotes

2 [Foundational] • Target: Standard form

Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.



3 [Foundational] • Target: Standard form

Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

a Write the equation in standard form by completing the square.

b Identify the conic and state its center.

4 [Foundational] • Target: Identifying conics

Identify each conic section from its equation:

a $x^2 + y^2 = 49$

b $\frac{x^2}{9} + \frac{y^2}{25} = 1$

c $y^2 - 4x^2 = 16$

d $y = 3x^2 - 6x + 1$

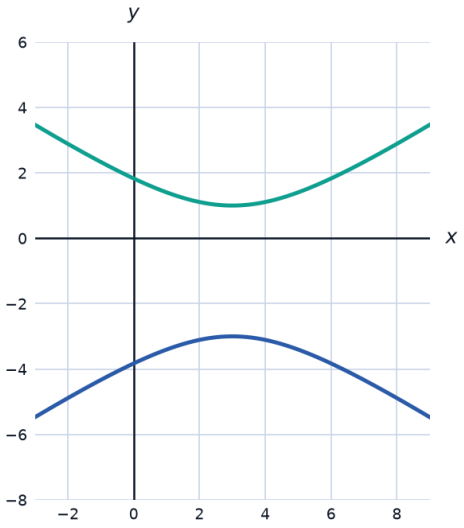
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5 [Foundational] • Target: Identifying conics

A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.

6 [Foundational] • Target: Hyperbolas centered away from the origin

When analyzing the behavior of the model, for the hyperbola $\frac{(y+1)^2}{4} - \frac{(x-3)^2}{9} = 1$, find:



a the center **b** the vertices **c** the direction of opening

7 [Foundational] • Target: Eccentricity and applications

Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

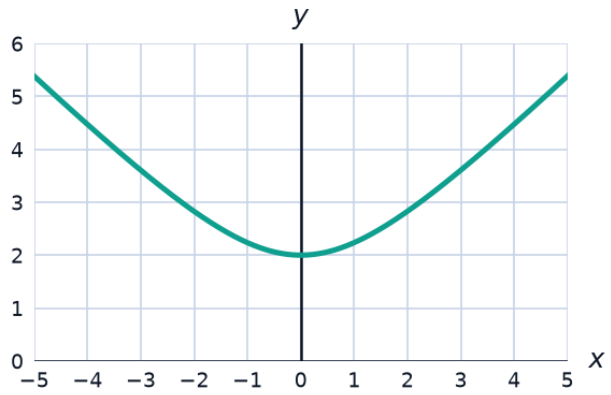
8 [Intermediate] • Target: Eccentricity and applications

A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

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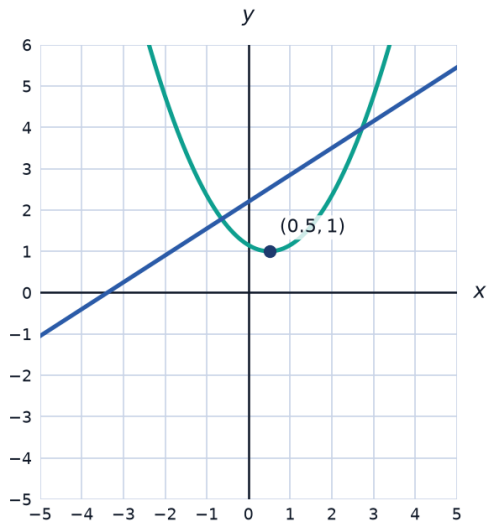
9 **[Intermediate]** • **Target: Graphing a hyperbola branch**

The upper-right branch of $y^2 - x^2 = 4$ (as $y = \sqrt{4 + x^2}$) is shown. Use the graph to estimate y at $x = 3$, then verify algebraically, and compare to the asymptote $y = x$.



10 **[Intermediate]** • **Target: The hyperbola's defining property**

State the geometric (difference-of-distances) definition of a hyperbola, and use it to verify the vertex $(4, 0)$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) satisfies it.



11 **[Intermediate]** • **Target: Comparing all four conic sections**

State the key algebraic feature (coefficient signs) that distinguishes a circle, ellipse, parabola, and hyperbola in a general second-degree equation $Ax^2 + By^2 + \ldots = 0$.

12 **[Intermediate]** • **Target: Finding the equation from asymptotes and a point**

A hyperbola centered at the origin has asymptotes $y = \pm\frac{2}{3}x$ and passes through $(3, 0)$. Find its equation.

13 **[Intermediate]** • **Target: Applications: LORAN navigation**

Two LORAN radio towers are 260 km apart. A ship determines the difference in distances to the two towers is 100 km. Write the equation of the hyperbola of possible ship locations (centered at the midpoint, opening along the line connecting the towers).

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14 **[Intermediate]** • **Target: Graphing a hyperbola branch**

Write the general circle equation $x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, identify its center and radius, and find the equation of the tangent line to the circle at the point $P(4, 1)$.

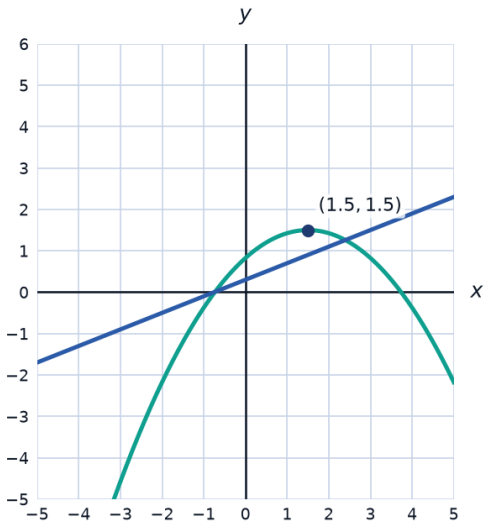
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The hyperbola's defining property** • **Cross-Lesson: 9.03 The Binomial Theorem**

Write $9x^2 + 4y^2 - 36x + 8y + 4 = 0$ in standard form and identify the conic.

16 **[Challenging]** • **Target: Comparing all four conic sections** • **Cross-Lesson: 1.01 Functions and Function Notation**

Determine the type of conic section represented by the general quadratic equation $4x^2 - 9y^2 + 16x + 54y - 101 = 0$, and find the coordinates of its center.



17 **[Challenging]** • **Target: Finding the equation from asymptotes and a point** • **Cross-Lesson: 1.02 Domain and Range**

Two chords AB and CD of a circle intersect at interior point M. If $AM = 4$ cm, $MB = 9$ cm, and $CM = 6$ cm, determine the length of MD.

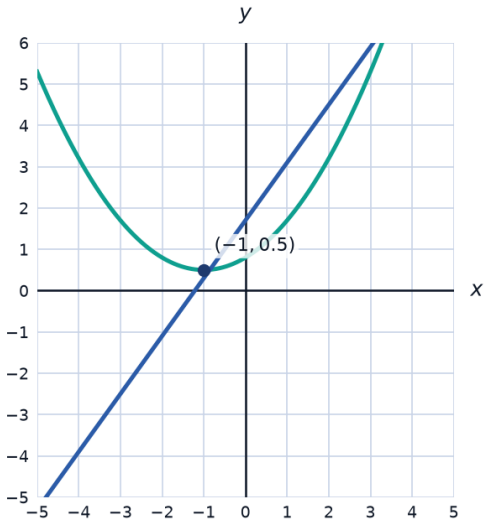
18 **[Challenging]** • **Target: Applications: LORAN navigation** • **Cross-Lesson: 1.03 Average Rate of Change**

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

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19 [Challenging] • Target: Standard form • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A polynomial $f(x)$ of degree 4 with real coefficients has leading coefficient 2 and zeros at $x = -1$ (multiplicity 2) and $x = 2 + i$. Write $f(x)$ in fully expanded standard form.



20 [Challenging] • Target: Identifying conics • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard form

Find the equation of the hyperbola with asymptotes $y = \pm 2x$ and passing through $(3, 4)$.

a. $\frac{x^2}{5} - \frac{y^2}{20} = 1$

b. $y^2 - 4x^2 = 0$

c. $4x^2 - y^2 = 32$

d. $x^2 - 4y^2 = 4$

22 [Foundational] • Target: Identifying conics

A hyperbola has vertices $(\pm 2, 0)$ and asymptotes $y = \pm 3x$. Find its equation.

a. $\frac{x^2}{4} - \frac{y^2}{9} = 1$

b. $\frac{x^2}{4} - \frac{y^2}{36} = 1$

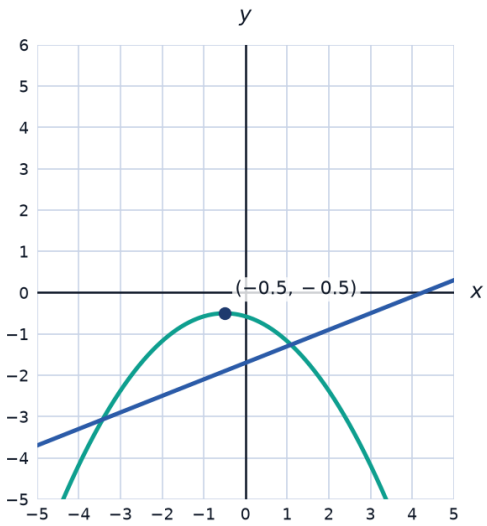
c. $y^2/9 - x^2/4 = 1$

d. $\frac{x^2}{9} - \frac{y^2}{4} = 1$

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23 [Foundational] • Target: Hyperbolas centered away from the origin

A hyperbola is defined by the equation $\frac{(x-h)^2}{16} - \frac{(y-k)^2}{9} = 1$. If the vertices of the hyperbola are located at $(5, 2)$ and $(-1, 2)$, determine the values of h and k.



a. h = 2, k = 2

b. h = 2, k = -1

c. h = 2, k = 5

d. h = 5, k = 2

24 [Foundational] • Target: Eccentricity and applications

Prove that the difference of the focal radii of a point on a hyperbola equals $\pm 2a$.

a. From definition $|PF_1 - PF_2| = 2a$, which defines the hyperbola

b. Only for $e > 2$

c. $PF_1 - PF_2 = 2c$

d. It equals $2b$

25 [Foundational] • Target: Graphing a hyperbola branch

Write the equation of the hyperbola with foci $(\pm 5, 0)$ and vertices $(\pm 3, 0)$.

a. $\frac{x^2}{9} - \frac{y^2}{16} = 1$

b. $x^2 - y^2 = 9$

c. $\frac{x^2}{16} - \frac{y^2}{9} = 1$

d. $\frac{x^2}{25} - \frac{y^2}{9} = 1$

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26 [Foundational] • Target: The hyperbola's defining property

The eccentricity of a hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is:

a. $\frac{4}{3}$

b. $\frac{5}{3}$

c. $\frac{3}{4}$

d. $\frac{3}{5}$

27 [Foundational] • Target: Comparing all four conic sections

Find the equation of the circle centered at $(0, 0)$ passing through $(3, 4)$.

a. $x^2 + y^2 = 7$

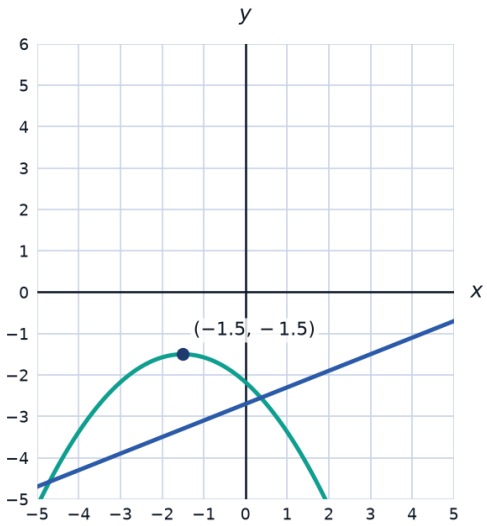
b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 12$

d. $x^2 + y^2 = 25$

28 [Foundational] • Target: Finding the equation from asymptotes and a point

For a hyperbola, the eccentricity satisfies:



a. $e > 1$

b. $e = 1$

c. $0 < e < 1$

d. $e = 0$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Applications: LORAN navigation

Find the angle between the asymptotes of the hyperbola $\frac{x^2}{4} - \frac{y^2}{4} = 1$.

a. 45°

b. 90°

c. 60°

d. 30°

30 [Intermediate] • Target: Standard form

The rectangular hyperbola $xy = c^2$ has eccentricity:

a. 1

b. 2

c. $\sqrt{2}$

d. $\sqrt{3}$

31 [Intermediate] • Target: Identifying conics

Find the equation of a sphere in $\mathbb{R}^3$ centered at $(1,\ -2, 2)$ with radius 3.

a. $x^2 + y^2 + z^2 = 9$

b. $(x-1)^2 + (y+2)^2 + (z-2)^2 = 3$

c. $(x+1)^2 + (y-2)^2 + (z+2)^2 = 9$

d. $(x-1)^2 + (y+2)^2 + (z-2)^2 = 9$

32 [Intermediate] • Target: Hyperbolas centered away from the origin

Find the equation of the hyperbola with transverse axis along y, $a = 3$, $b = 4$.

a. $\frac{y^2}{16} - \frac{x^2}{9} = 1$

b. $\frac{x^2}{9} - \frac{y^2}{16} = 1$

c. $\frac{y^2}{9} - \frac{x^2}{16} = 1$

d. $y^2 - x^2 = 3$

33 [Intermediate] • Target: Eccentricity and applications

Find the equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(a\cos\theta, b\sin\theta)$.

a. $x = a\cos\theta$

b. $x\cos\theta/a + y\sin\theta/b = 1$

c. Normal cannot be found in closed form

d. $ax\sec\theta - by\csc\theta = a^2 - b^2$

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34 [Intermediate] • Target: Graphing a hyperbola branch

A hyperbola has $a = 4$, $b = 3$. Find the equations of the asymptotes.

a. $y = \pm 4x$

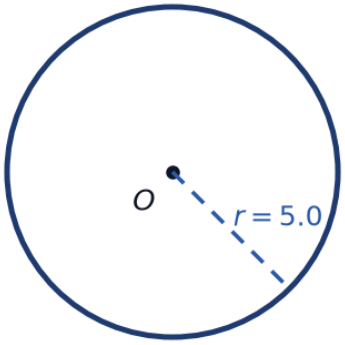
b. $y = \pm 3x$

c. $y = \pm \frac{3}{4}x$

d. $y = \pm \frac{4}{3}x$

35 [Intermediate] • Target: The hyperbola's defining property

Find the equation of the circle with center $(3, -1)$ and radius 5.



a. $(x + 3)^2 + (y - 1)^2 = 25$

b. $(x - 3)^2 + (y + 1)^2 = 25$

c. $(x - 3)^2 + (y - 1)^2 = 5$

d. $x^2 + y^2 = 25$

36 [Intermediate] • Target: Comparing all four conic sections

The equation of an ellipse is given by: $9x^2 + 4y^2 - 36x + 8y + k = 0$ If the length of the major axis of the ellipse is 6, what is the value of the constant k?

a. 4

b. -31

c. -41

d. -104

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37 [Intermediate] • Target: Finding the equation from asymptotes and a point

An ellipse in the xy-plane is defined by the equation $9x^2 + 16y^2 - 54x + 64y + c = 0$ where c is a constant. If the length of the minor axis of the ellipse is 6, what is the value of c?

a. 1

b. -1

c. 64

d. 289

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications: LORAN navigation • Cross-Lesson: 1.03 Average Rate of Change

The function f is defined by $f(x) = \frac{2x^2 + kx - 6}{x - 2}$ for all real numbers $xeq2$, where k is a constant. If the graph of $y = f(x)$ in the standard (x, y) coordinate plane has a removable discontinuity (a hole) at the point $(2, L)$, what is the value of L?

a. -1

b. 3

c. 7

d. 11

39 [Challenging] • Target: Standard form • Cross-Lesson: 2.01 Right-Triangle Trigonometry

In the diagram below, chords AB and CD intersect at point P. If AP = 6, PB = 8, and CP = 4, find the length of PD.

a. 3

b. 4.5

c. 6

d. 12

40 [Challenging] • Target: Identifying conics • Cross-Lesson: 2.02 Radian Measure and Arc Length

What are the coordinates of the center and the radius of the circle given by the equation $(x - 4)^2 + (y + 1)^2 = 36$?

a. Center $(4, -1)$; radius 6

b. Center $(-4, 1)$; radius 6

c. Center $(4, -1)$; radius 36

d. Center $(-4, 1)$; radius 36

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41 [Challenging] • Target: Hyperbolas centered away from the origin • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

What are the center and radius of the circle given by the equation $(x-2)^2+(y+6)^2=36$?

a. Center $(2,-6)$, radius 6

b. Center $(-2,6)$, radius 6

c. Center $(2,-6)$, radius 36

d. Center $(-2,6)$, radius 36

42 [Challenging] • Target: Eccentricity and applications • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

What is the center of the circle represented by the equation $(x-3)^2+(y+5)^2=16$?

a. (3, 5)

b. (3, -5)

c. (-3, 5)

d. (-3, -5)

43 [Challenging] • Target: Graphing a hyperbola branch • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Prove that the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on $y^2=4ax$ has equation $y(t_1+t_2)=2x+2at_1t_2$.

a. Use calculus

b. Incorrect formula

c. Slope $=2/(t_1+t_2)$; using point-slope through $(at_1^2, 2at_1)$ gives the result

d. Only for $t_1=t_2$

44 [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Derive the eccentricity of a conic from the focus-directrix ratio definition $PF/PD=e$ for the parabola.

a. $e=2$ for parabola

b. For parabola, $e=1$: distance to focus equals distance to directrix

c. $e>1$ for parabola

d. $e<1$ always

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45 [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the equation of the ellipse with foci $(\pm 3, 0)$ and sum of focal radii $= 10$.

a. $\frac{x^2}{10} + \frac{y^2}{9} = 1$

b. $\frac{x^2}{9} + \frac{y^2}{16} = 1$

c. $\frac{x^2}{25} + \frac{y^2}{16} = 1$

d. $x^2/16 + y^2/9 = 1$

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